

Model paper 2

Part 5

Question number 1

Consider the following trees and the statements given below:

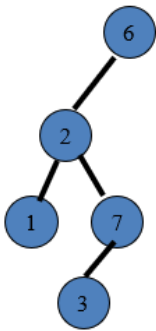


Figure 1

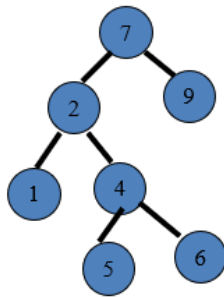


Figure 2

- (I) Both Figure 1 and Figure 2 are Binary Trees
- (II) Figure 1 is a Binary Search Tree
- (III) Figure 2 is a Binary Search Tree
- (IV) Both Figure 1 and Figure 2 are Binary search trees and not Binary Trees

Which of the following is correct?

- a) (I) only
- b) (I) and (II) only
- c) (I), (II) and (III) only
- d) ALL (I), (II), (III) AND (IV)

Question Number 2

One creates the Binary search tree and he/she wants to print in increasing order of nodes in the tree.

Which traversal techniques would be best for you to do the task above?

- a) Pre-order traversal
- b) Post-order traversal
- c) In-order traversal
- d) Breadth First Traversal

Correct answer c:

Question 3

The following numbers are inserted into an empty binary search tree in the given order:

13, 7, 3, 5, 20, 12, 17, 26, 30, 6

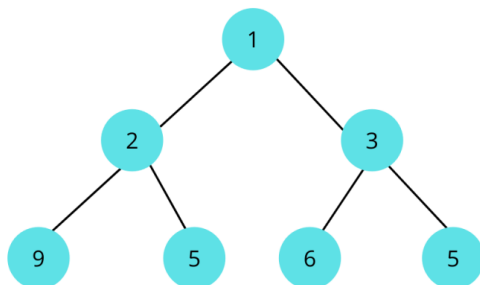
What is the height of the binary search tree (the height is the maximum distance of a leaf node from the root)?

- a) 1
- b) 2
- c) 3
- d) 4

Correct answer d

Question number 4

Consider the following Binary Tree.



What is the correct order of nodes of the tree if traversed in the following sequence?

Pre-order
In-order
Post-order
Breadth First Order

(a) (1,2,9,5,3,6,5), (9,2,5,1,5,3,6), (9,5,2,6,5,3,1), (1,2,3,9,5,6,5)

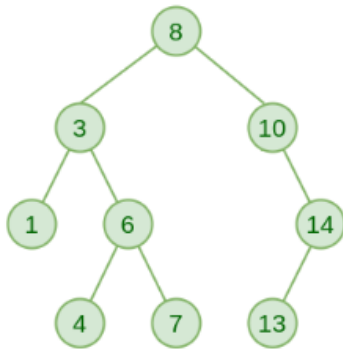
(b) (1,2,9,5,3,6,5), (9,2,5,1,6,3,5), (9,5,2,6,5,3,1), (1,2,3,9,5,6,5)

(c) (1,2,9,5,3,6,5), (9,2,5,1,6,3,5), (2,9,5,6,5,3,1), (1,2,3,9,5,5,5)

(1,2,5,9,3,6,5), (9,2,5,1,6,3,5), (9,5,2,6,5,3,1), (1,2,3,9,5,5,5)

Correct answer b)

Question 5



Suggest (a) suitable node(s) as the replacing node(s), if one wants to delete the node with the key value 8 from the above binary search tree.

- a) 3 and 10
- b) 7 only
- c) 6 and 14
- d) 7 and 10

Correct answer d)

Question 6)

Consider the following algorithm with blanks, which can be used to insert a new node into the binary search tree:

Insert (tree, VAL)

Step 1 : if tree=null

 Allocate memory for tree

 Set tree->data=VAL

 Set tree->left=tree->right=.....(i).....

else

 If VAL < tree->data

 Insert (...(ii).....), VAL

 else

 insert (.....(ii).....),VAL

[end of if]
 [end of if]
Step 3: end

The most suitable terms for the above blanks are:

- a) (i)null, (ii) t->left (iii) t->right
- b) (i)null, (ii) t->right (iii) t->left
- c) (i)root, (ii) t->left (iii) t->right
- d) (i)root, (ii) t->right (iii) t->left

Correct answer a)

Question number 7

Consider the following pseudocode algorithm segment

```
If (t = null)
return null;
else if (t->left = null)
return t;
return pqr(t->left);
```

what is the above algorithm segment intended to do?

- a) Count the number of nodes in a binary search tree
- b) Find the maximum value from a Binary search tree
- c) Find the minimum value from a binary search tree
- d) Calculate the height of a tree

Correct answer c)

Question number 8

The following array is given.

[1,2,4,8,5,3,7]

Bubble sort is used to sort the above array. How many iterations will be done to sort the array?

- a) 0
- b) 5
- c) 6
- d) 7

Correct answer c)

Question number 9

Consider the following pseudocode algorithm

```
abc (data[]){  
  for (i=0;i<data.length-1;i++)  
    temp=data[i];  
  Move all elements data[i] greater than temp by one position;  
  Place temp in its proper position;
```

what does the above algorithm best describe?

- a) Insertion sort
- b) Bubble sort
- c) Selection sort
- d) Quick sort

Correct answer a)

Question 10

Consider the following statements

- I. Set min to location 0
- II. Repeat until the array is sorted
- III. Increment min to point to next element
- IV. Swap with the value at location min
- V. Search the minimum element in the array.

Which order of the statements best describes the selection sort?

- a) (I), (IV), (V), (III), (II)
- b) (I), (V), (IV), (III), (II)
- c) (I), (V), (III), (IV), (II)
- d) (I), (V), (IV), (II), (III)

Correct answer b)

Part 6

Question 1

Sort the following array using selection sort. How many iterations will be done to sort the array?
[43,2,52,6,63,21,12]

- a) 3 Only
- b) 4 Only
- c) 5 Only
- d) 6 Only

Answer b

Question 2

Following array of nine elements is given to be sorted using quick sort.

[54,26,93,17,77,31,44,55,20]

The algorithm chooses the first element as its pivot. If only the first partitioning has been completed, which of the following shows the correct order of elements?

- a) 17,20,26,31,44,54,55,77,93
- b) 26,17,31,44,20,54,93,77,55
- c) 26,17,31,44,20,54,93,55,77
- d) 26,31,17,44,20,54,93,77,55

Correct answer b

Question 3

Which of the following statements is true about the Bubble Sort algorithm?

- a) Bubble Sort is a divide-and-conquer algorithm.
- b) Bubble Sort has a worst-case time complexity of $O(n \log n)$.
- c) Bubble Sort is generally more efficient than Quick Sort for large datasets.
- d) Bubble Sort compares each pair of adjacent elements and swaps them if they are in the wrong order.

Correct d)

Question 4

There is an unsorted array as follows having 6 elements. The array needs to be sorted in ascending order. Heap sort is used to sort the array. Which of the following statement is correct?

[4,3,7,1,8,5]

- a) During the creation of the maximum heap, 7 is swapped with 5
- b) During the creation of the maximum heap, 8 is swapped with 3
- c) During the creation of the maximum heap, 1 is swapped with 3
- d) During the creation of the maximum heap, 4 is swapped with 7

Correct answer c)

Question 5

Which of the following is true related to the quick sort algorithm?

- a) Select a pivot element from the array. This can be the first element, the last element, a random element, or the median.
- b) Root value is higher with respective all values in left sub-tree
- c) Quick sort is improved version of the insertion sort
- d) All values in the left sub-array is higher than the pivot value and all values in the right-sub array is less than or equal to the pivot value

Correct answer a)

Question 6

Which of the following is **incorrect** related to the shell sort algorithm?

- a) If the increment value (k) is 5, then the original file can be divided in to 5 sub-files.
- b) In general, I^{th} element of the J^{th} sub file is $x[(i-1)*k^j]$
- c) A series of increments need to sort the entire file
- d) Shell-sort is faster than the straight insertion sort

Correct answer is b)

Question 7

Consider the following pseudocode algorithm segment?

```
return = A[0];  
A[0] = A[N-1];  
N--;  
Upheap[N];
```

This algorithm can be used to:

- a) delete a node from a heap
- b) insert a node to the heap
- c) remove the largest element from the queue
- d) change the priority of a node in a queue

correct answer c)

Question 8

Which of the following statements is true about the selection sort algorithm?

- a) Selection sort is an in-place sorting algorithm.
- b) the selection sort finds the smallest element in the array and exchanges it with middle element in the array.
- c) Selection sort does not require additional memory proportional to the size of the input array; it only requires a constant amount of extra space.
- d) Sorting mechanism is sorting by insertion

Correct answer a)

Question Number 9

A list of 7 numbers are rearranged in ascending order using a bubble sort algorithm. Maximum number of swaps were used. What can be deduced about the original list of numbers?

- (a) Original data is in ascending order
- (b) Original data is in descending order
- (c) Original data set is in random order
- (d) Original data set is unordered

Correct answer b)

Question 10

Which of the following sorting algorithms is **not** a divide-and-conquer type?

- a) Merge sort
- b) Quick Sort
- c) Shell Sort
- d) Insertion sort

Correct answer d)